Aodong Liu

Computational Chemistry | Nuclear Quantum Effects | Scientific HPC

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Research Profile

Computational chemist developing nuclear–electronic orbital theory, scalable electronic-structure algorithms, and coupled nuclear/electronic quantum dynamics. Lead NEO-module developer for Gaussian 25 and active scientific-software developer in Fortran and C++, with nine peer-reviewed journal articles (first-listed author on three and co-first author on a fourth) and extensive MPI/OpenMP/HPC experience.

Appointments

University of Washington — **Seattle, WA**
Postdoctoral Scholar, Li Group — *Aug. 2026–Present*

- Continue development of scalable quantum-chemistry and coupled nuclear/electronic dynamics methods.

University of Washington — **Seattle, WA**
Graduate Research Assistant, Li Group — *Sep. 2021–Jun. 2026*

- Developed NEO electronic-structure and non-Born–Oppenheimer dynamics methods and their scalable scientific-software implementations.
- Co-developed a distributed Cholesky framework demonstrated on 52 AMD EPYC nodes/104 MPI ranks (9,984 bound CPU cores), enabling a 500-quantum-proton calculation and 96.2-second full NEO Fock build.

Gaussian, Inc. — **Wallingford, CT**
Software Developer — *Feb. 2023; Apr. and Sep. 2024*

- Led development of the Nuclear–Electronic Orbital module in Gaussian 25.

University of Washington — **Seattle, WA**
Teaching Assistant, Graduate Computational Chemistry (CHEM 465) — *Winter 2022, 2023*

Education

University of Washington — **Seattle, WA**
Ph.D., Theoretical Chemistry; Advisor: Xiaosong Li — *Jun. 2026*
Certificate in Quantum Information Science and Engineering.

Vanderbilt University — **Nashville, TN**
B.A., Physical Chemistry with Honors; Minor, Scientific Computing — *May 2020*
Undergraduate advisor: Jens Meiler. Thesis: *de novo* modeling of CNIH1 and its docking mechanism with the AMPA receptor.

Technical Expertise

Programming: C++, Fortran, Python, Bash/Shell; C, CUDA, Java, MATLAB.
HPC/software: MPI, OpenMP, distributed-memory algorithms, CMake, Git, Linux, CI/CD, Gaussian, ChronusQ.
Methods: NEO theory, electronic structure, DFT, Cholesky decomposition, relativistic methods, real-time and quasiclassical dynamics.

Peer-Reviewed Publications

1. Zhang, T.; **Liu, A.**; Upadhyay, S.; Hu, H.; Beck, R. A.; Bersson, J.; Boyer, N.; Culpitt, T.; Garner, S.; Kovtun, M.; Li, R. R.; Liao, C.; Lu, L.; Majumder, R.; Oele, M.; Peltekis, A. A.; Petrone, A.; Shayit, A.; Sun, S.; Tang, D.; Wang, T.; Wildman, A.; Williams-Young, D.; Yuwono, S. H.; Zhao, L.; Lambros, E.; Hoyer, C. E.; Valeev, E. F.; DePrince, A. E., III; Hammes-Schiffer, S.; Li, X. "Chronus Quantum 1.0: A Unified Framework for Relativistic, Time-Dependent, and Multicomponent Electronic Structure Theory." *APL Computational Physics* **2**, 046102 (2026). doi:10.1063/5.0356458.

2. Tang, D.; **Liu, A.**; Culpitt, T.; Li, X. "Ab Initio Two-Spinor Ehrenfest Dynamics with Magnetic Field." *Phys. Chem. Chem. Phys.* **28**, 19461–19471 (2026). doi:10.1039/D6CP02263D.

3. Yang, M.; Upadhyay, S.; **Liu, A.**; Li, X. "Spin–Orbit-Resolved Strong-Field Ionization from Real-Time Relativistic Dynamics." *J. Chem. Phys.* **165**, 014105 (2026). doi:10.1063/5.0320154.

4. **Liu, A.**; Zhang, T.; Lambros, E.; Hammes-Schiffer, S.; Li, X. "Nuclear–Electronic Orbital Method at Scale: A Two-Stage Hybrid-Layout Distributed Cholesky Decomposition Approach." *APL Computational Physics* **2**, 026109 (2026). Editor's Pick. doi:10.1063/5.0318199.

5. Aieta, C.; Garner, S. M.; **Liu, A.**; Li, X.; Hammes-Schiffer, S. "Nuclear–Electronic Orbital Quasiclassical Trajectory Method for Vibrational Spectroscopy." *J. Chem. Phys.* **164**, 144123 (2026). doi:10.1063/5.0317500.

Peer-Reviewed Publications (continued)

6. Tang, D.*; **Liu, A.***; Culpitt, T.; Hammes-Schiffer, S.; Li, X. "Simulating Magnetic Field-Driven Real-Time Quantum Dynamics Using London Nuclear–Electronic Orbital Approach." *J. Chem. Theory Comput.* **21**, 4357–4364 (2025). doi:10.1021/acs.jctc.5c00273.

7. Kovtun, M.; Lambros, E.; **Liu, A.**; Tang, D.; Williams-Young, D. B.; Li, X. "Accelerating Relativistic Exact-Two-Component Density Functional Theory Calculations with Graphical Processing Units." *J. Chem. Theory Comput.* **20**, 7694–7699 (2024). doi:10.1021/acs.jctc.4c00843.

8. **Liu, A.**; Zhang, T.; Hammes-Schiffer, S.; Li, X. "Multicomponent Cholesky Decomposition: Application to Nuclear–Electronic Orbital Theory." *J. Chem. Theory Comput.* **19**, 6255–6262 (2023). doi:10.1021/acs.jctc.3c00686.

9. **Liu, A.**; Chow, M.; Wildman, A.; Frisch, M. J.; Hammes-Schiffer, S.; Li, X. "Simultaneous Optimization of Nuclear–Electronic Orbitals." *J. Phys. Chem. A* **126**, 7033–7039 (2022). doi:10.1021/acs.jpca.2c05172.

*Equal contribution.

Pending Manuscripts

1. **Liu, A.**; Beck, R.; Upadhyay, S.; Garner, S. M.; Graham, T.; Ghandi, K.; Pearce, C.; Hammes-Schiffer, S.; Li, X. "Vibrational Bonding in Brμ·Br from Correlated Muonic–Electronic Wavefunctions." Under revision, 2026.

2. **Liu, A.**; Lambros, E.; Beck, R.; Hammes-Schiffer, S.; Li, X. "Forming the Bound Electron: Non-Equilibrium Electronic Ionization Dynamics of Water Clusters." In preparation, 2026.

3. Zheng, J.; **Liu, A.**; Frisch, M. J. "A Tutorial for Nuclear and Electron Dynamics in the Gaussian Program." Submitted, 2026.

Selected Presentations

- ACS Spring 2026, oral: "Large-Scale Real-Time NEO–TDDFT Simulations of Ultrafast Solvated Electron Dynamics," Atlanta, March 2026.
- IDREAM EFRC All Hands Meeting, PNNL, October 2025.
- IDREAM EFRC Meeting, oral: "Ultrafast Processes After Low-Energy Electron Attachment in Water Clusters," PNNL, October 2024.
- ACS Spring 2024, oral: "Multicomponent Cholesky Decomposition: Application to Nuclear–Electronic Orbital Theory," New Orleans, March 2024.
- EFRC–Hub–CMS–CCS PI Meeting, poster: "Multiscale Nuclear–Electronic Orbital Quantum Dynamics in Complex Environments," virtual, September 2023.
- EFRC–Hub–CMS–CCS PI Meeting, poster: "Radiolysis in Extreme Environments," virtual, September 2023.
- IDREAM EFRC Meeting, oral: "A Computational Study of Potassium K-Edge XANES Spectra," PNNL, August 2023.
- SciDAC Summit, oral: "Simultaneous Optimization of Nuclear–Electronic Orbitals," LBNL, December 2022.
- Vanderbilt Student Research Fair, oral: "Structural Modeling and Validation of CNIH1 Protein," Nashville, September 2019.

Awards

- Excellence in Chemistry Graduate Fellowship Award, University of Washington, 2021.
- Thomas W. Martin Award, Vanderbilt University, 2020; Dr. Sidney & Becca Fleischer Award, 2019.
- Vanderbilt Undergraduate Summer Research Program Fellowship, 2018.

Leadership and Service

- Hyak Governance Board member and Research Computing Club officer, University of Washington, since 2023; RCC mentor since 2022.
- Engineering Discovery Days volunteer: developed and delivered hands-on K–12 demonstrations of superconductivity, quantum levitation, and MAGLEV principles; guided visitors through the underlying physical concepts.